

Sampling II: Non-Probability Sampling

Wenchao Ma

University of Minnesota



Quota Sampling

- Similar to stratified sampling in that we develop non-overlapping groups
- Sample a predetermined number of individuals within each group
- An example:

Age Group	Distribution in Population	Quota to Achieve n=300
<20	30%	n=90
20-49	40%	n=120
50+	30%	n=90



Quota Sampling

- Not the same as a stratified sample
 - Still a non-probability sample
 - Improves variation within the sample to reflect some aspect(s) of the population
 - Still subject to selection bias & subjectivity



Purposive Sampling

- Also called **judgmental sampling**
- Participants chosen based on their specialized knowledge of the subject area



Purposive Sampling

- Not a probability sample
- Sample drawn is not representative of the larger population
 - Cannot estimate sampling error or bias



An Example of Purposive Sampling: Extreme case sampling

- focus on cases that are special or unusual, typically in the sense that the cases highlight notable outcomes, failures or successes.

A researcher wants to collect feedback from students on a new instructional plan.

She pick some students who are less satisfied to gain more insights.



Convenience Samples

- Elements selected because they are easy to include, or they are available
- Common problem:
 - Respondents will have similar characteristics because they are part of the researcher's network



Convenience Samples

- Least-scientific form of sampling
- Other names:
 - Availability sampling
 - Haphazard sampling



Convenience Samples

- Interpret results with extreme caution
- No reason to believe the sample is representative of the population



Sampling Hard-to-Find or Hidden Populations

- Some populations are hard to sample
 - Size & locations are not known
 - Population is marginalized or stigmatized
- Examples?
 - Homeless
 - drug users
 - Others?

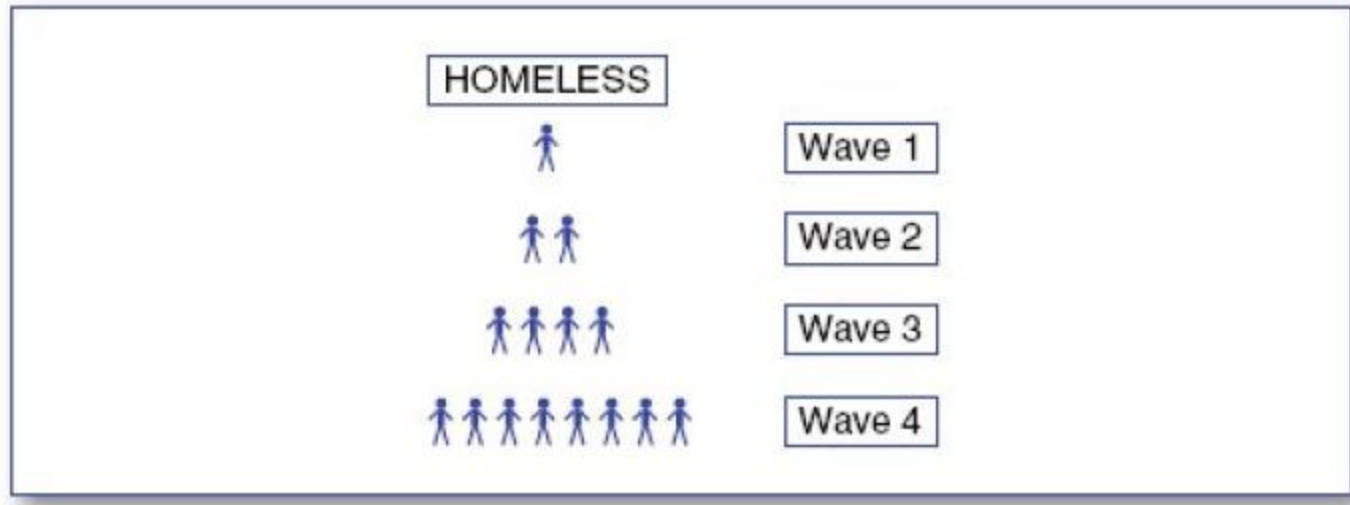


Snowball Sampling

- Other names:
 - Chain sampling
 - Chain referral
- A member of the sample refers the researcher to others who fall within the population
 - These members refer others
 - These members refer others
 - And so on...



Snowball Sampling



What are problems that come to mind with snowball sampling?



Selection Bias

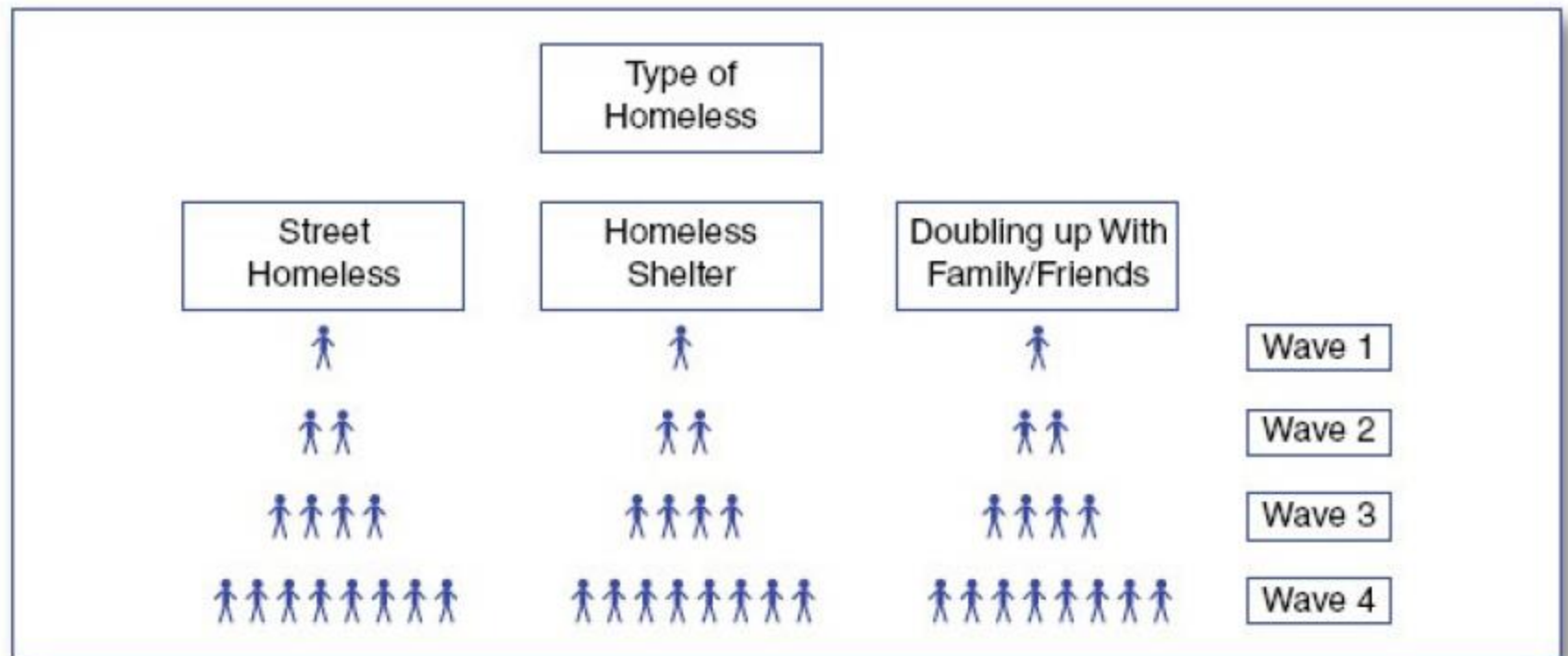
- Whoever is included as the Wave 1 recruit can have a strong influence on the rest of the sample



Minimizing bias in snowball sampling

- Use Wave 1 recruits from different sub-populations
- Draw Wave 1 recruits randomly if possible





Exercise: Which sampling method is used?

- A researcher wants to conduct a survey to study the eating habits of UMN students.
 - They randomly select 100 students from the list of all enrolled students at the university.
 - They randomly select four from eight freshman-admitting colleges and then recruit students by advertising at the college buildings.
 - They go to St. Paul Gym to find students who are interested in this survey.
 - They go to Dinky town to identify 25 freshman, 25 sophomore, 25 junior and 25 senior undergraduates.



Exercise: Which sampling method is used?

- A researcher studying a rare disease starts with a few known patients and asks them to refer other patients they know.
- A student conducts a survey on study habits by asking friends and classmates to participate.
- A researcher conducts a survey of attitudes of drug users to rehabilitation services. Drug users are recruited by advertising in the local newspaper for potential respondents.



Exercise: Which sampling method is used?

- A survey of the attitudes of mothers with children under one year.
 - Method: interviewers stop likely looking women pushing prams in the street. The number of respondents who fall into different age bands and social classes is strictly controlled.
- A survey for children with and without chronic asthma.
 - Method: the children are selected so that in the sample exactly 50% have chronic asthma and 50% have no asthma. Within each of these groups, the children are randomly selected.

